AAKASH GURUMURTHI

B.Tech Information Technology Student | Full-Stack & ML Developer

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PROFESSIONAL SUMMARY

Final-year Information Technology student with hands-on development experience across machine learning, cybersecurity, distributed systems, and full-stack engineering.

TECHNICAL SKILLS

Programming Languages: Python, C++, C, Java, SQL, PHP
Web & Mobile Development: HTML5, CSS3, JavaScript (ES6+), React.js, Node.js, Flutter, FastAPI, Flask
Machine Learning & AI: HuBERT, Random Forest, LSTM, Autoencoders, Vision/Swin Transformers, Computer Vision, Librosa
Cybersecurity & Networking: Snort 3, Intrusion Detection Systems, UDP/TCP Protocols, VirusTotal API
Databases & Tools: MySQL, MongoDB, SQLite, Firebase, Git/GitHub, Android Studio

EDUCATION

Vellore Institute of Technology, Vellore, Tamil Nadu *August 2023 - Present*
B.Tech in Information Technology | CGPA: 8.54
Relevant Coursework: Database Systems, Operating Systems, Data Structures & Algorithms, AI & ML

Seth Anandram Jaipuria School, Vasundhara, Ghaziabad *March 2022 - March 2023*
Class 12, Science (PCM) | CBSE Board | Score: 92%

Seth Anandram Jaipuria School, Vasundhara, Ghaziabad *March 2020 - March 2021*
Class 10 | CBSE Board | Score: 97.5%

PROFESSIONAL EXPERIENCE

App Development Intern, Centre for Railway Information Systems (CRIS) *June 2025 - July 2025*
- Developed native user interface pages for the IRCTC mobile application using Flutter and Android Studio.
- Built a cross-platform Events Planner application integrated with secure Firebase Authentication workflows.

Web Development and Database Intern, DLF Ltd. *May 2025 - June 2025*
- Queried complex commercial datasets in a case study about a distributor of mobile phones using relational joins and aggregate functions on MySQL.
- Built an interactive data dashboard using HTML, CSS, JavaScript, PHP, and SQL to track leasing metrics.

KEY PROJECTS

Drift-Aware Hybrid Intrusion Detection System (DAHIDS)
- Engineered a production-ready hybrid intrusion detection system merging Snort 3 signature matching with an Adaptive Random Forest, LSTM, and Autoencoder model triad to detect zero-day threats.
- Designed 7 custom self-healing workflows, including Time-Machine Rollback and Confidence Gating, reducing false-positive rates to under 3%.
- Structured a FastAPI backend processing 100+ network flows per second with sub-50ms processing latency.
- Delivered a real-time React/MUI dashboard via WebSockets for live traffic visualization, model health monitoring, and custom Snort rule recommendations.

Decentralized Peer-to-Peer Messaging System - Patented (Application No. 202641066395, India, Filed May 2026)
- Designed a zero-configuration, fault-tolerant decentralized messaging network in Python using UDP multicast for node discovery.
- Implemented gossip protocol and anti-entropy mechanisms over TCP for eventual consistency, with Lamport logical clocks for absolute event ordering.

- Managed local message stores using SQLite and built a real-time Flask web dashboard visualizing message streams, network topology, and peer status.

Multi-Lingual Sign Language Translation System

- Built a computer vision pipeline utilizing a Swin/Vision Transformer architecture to translate sign language gestures into text across multiple languages.
- Managed real-time camera frame streams using OpenCV to achieve low-latency gesture recognition and translation.

Speech Emotion Recognition for Opera Performances

- Extracted audio waveforms processed at 16kHz via Librosa using the HuBERT Base framework for feature extraction.
- Trained a Random Forest classifier that reliably identifies operatic vocal sentiments from uploaded audio files.

PhishBreaker: Machine Learning URL Analyzer

- Developed an endpoint security utility classifying malicious URLs based on 30 structural and network features.
- Integrated the VirusTotal API to supply live domain reputation intelligence to a Flask web application interface.

AI-Based Medical Report Analyzer

- Built a tool enabling users to upload X-rays and pathology reports to receive anomaly inference, severity scoring, and precautions in multiple regional languages via a React-based dashboard.

HONORS & LEADERSHIP

- SSB Recommended Candidate: AIR 462 (NDA-151 Course), 2023.
- Best Electronics Team: Awarded first place at the Yantra Central Hackathon for an AI-based automated fruit sorting system, controllable via a mobile application.
- Head of Events, RoboVITics Robotics Club (2025–2026): Managed operations, budgets, and technical logistics for university-wide events. Also serving in Advisory role (2026–2027) and as App Developer (2024–2025).